

Dynamic anionic sublattice engineering in perovskite heterostructures for perovskite–silicon tandem solar cells

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Wide-bandgap perovskite top cells for tandem photovoltaics are constrained by voltage deficit and non-radiative losses. Here we design a series of perovskite derivatives featuring dynamically disordered cyanate anions and demonstrate that the dynamic anionic sublattice engineering alters crystallization kinetics, directing the epitaxial growth of a coherent bulk heterojunction with large-grained morphology. The cyanate anions provide chemical and electronic passivation at grain boundaries and interfaces, neutralizing deep-level traps and suppressing non-radiative recombination. The single-junction wide-bandgap perovskite solar cell fabricated with this method has a power conversion efficiency of 24.35%. When integrated as the top cell in a monolithic perovskite–silicon tandem, a power conversion efficiency of 34.79% (certified 34.29%) is achieved. Exceptional operational stability is demonstrated, with 95% initial performance retained after 1,100 hours of continuous illumination under the ISOS-L-1 protocol. Extended real-world testing over 1,300 hours under the ISOS-O-2 protocol further demonstrates its enhanced stability. This work establishes the rational design of perovskite anionic synthesis chemistry as a powerful route to overcome the intrinsic stability–performance trade-off in perovskite–silicon tandem cells.

Metal halide perovskites have redefined the landscape of next-generation photovoltaics, with single-junction cell efficiencies now rivaling those of established crystalline silicon technology^{1,2}. This rapid progress extends to tandem architectures, where perovskite–silicon stacks have shown the potential to surpass the theoretical efficiency limits of single-junction devices, recently achieving certified efficiencies exceeding 34% (refs. 3–5). However, the path to commercialization is obstructed by persistent and interrelated fundamental challenges, particularly for wide-bandgap (>1.68 eV) perovskite

top cells essential for high-performance tandem devices. First, the solution-processed crystallization of these materials remains difficult to control, typically resulting in films with small, randomly oriented grains, abundant grain boundaries and incomplete surface coverage. These microstructural imperfections act as nucleation sites for defect formation and pathways for ion migration^{6,7}. Second, while surface treatments can temporarily passivate interfacial traps, they often fail to address the bulk and grain-boundary defects that proliferate under operational stressors such as light, heat and electric fields⁸.

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Third, the most common strategies for widening the bandgap, such as increasing bromide content, exacerbate phase segregation and halide migration, leading to pronounced open-circuit voltage (V_{oc}) deficits and rapid photo-instability^{9,10}. Consequently, a persistent efficiency–stability trade-off limits the field, underscoring the limitation of current incremental modifications within the conventional halide perovskite framework^{11,12}.

To address defects, advanced passivation has moved beyond simple surface treatments to incorporate bulk additives and grain-boundary modifiers, such as potassium cations¹³, tribromide ions¹⁴ or low-dimensional perovskites^{15,16}, which provide more spatially homogeneous defect suppression. To address phase instability, compositional innovation via partial anion substitution (for example, with acetate or thiocyanate) or the use of stabilized one-dimensional (1D)/two-dimensional/three-dimensional (3D) heterostructures has shown promise in mitigating halide segregation under illumination^{17,18}. More recently, thiocyanate (SCN^-) and cyanate (OCN^-) anions have been used as external kinetic modulators for passivation and crystallization control^{19,20}. While these approaches have individually advanced device performance, they often operate in isolation. A strategy that governs crystallization kinetics, suppresses defects throughout the bulk and at interfaces and locks in the desired phase under operational conditions remains elusive. Most critically, these challenges are interdependent; addressing one often aggravates another. For instance, additives that improve crystallization may introduce charge-extraction barriers, whereas phase-stabilizing compositions may compromise charge transport^{21,22}. More fundamentally, the deeper structure–property relationship, specifically, how the integration of components fundamentally reconstitutes the electronic and ionic landscape of the perovskite lattice to intrinsically suppress instability, remains largely unexplored. These approaches, like their predecessors, operate within models of external compensation. They treat symptoms rather than addressing the root cause, failing to resolve the intrinsic performance–stability trade-off, as they do not fundamentally restructure the material's origin. What is critically missing is a solution that targets the material's origin, concurrently arresting its inherent ionic and structural dynamics to permanently lock in efficiency from the ground up.

Here we design and synthesize a series of perovskite heterostructures incorporating OCN^-/SCN^- -based low-dimensional perovskite (for example, $RbPbI_2OCN$, $MAPbI_2OCN$, $KPbI_2OCN$, $CsPbI_2OCN$, $RbPbI_2SCN$ and $MAPbI_2SCN$) via dynamic anionic sublattice engineering. The dynamically disordered OCN^- anions redefine crystallization pathways, enable the formation of a coherent bulk heterostructure with superior optoelectronic quality and provide built-in passivation. This single materials-level intervention resolves the core challenges of uncontrolled crystallization, bulk defect proliferation and ion migration. It enables a 24.35%-efficient wide-bandgap perovskite cell and, critically, a monolithic perovskite–silicon tandem solar cell (TSC) achieving a power conversion efficiency (PCE) of 34.79% (certified 34.29%) with exceptional operational stability.

Microstructure and electronic structure of $RbPbI_2OCN$

We designed and prepared perovskite single crystals using a solvent evaporation method by mixing $RbOCN$ into a lead-iodide (PbI_2) precursor solution. Our observations indicate that the presence of methylammonium chloride (MACl) substantially enhances the solubility of $RbOCN$ in dimethyl sulfoxide (DMSO). This effect originates from competition in dissolution thermodynamics between lattice dissociation energy and ionic solvation stability²³. MACl mitigates the aggregation tendency of $RbOCN$ species, thereby improving solution homogeneity and promoting a more uniform distribution of ionic constituents in the precursor system (Supplementary Fig. 1). Detailed instructions for single-crystal growth and X-ray diffraction test analysis are provided in

Methods (Supplementary Figs. 2 and 3). Energy-dispersive X-ray spectroscopy (EDS) mapping and X-ray photoelectron spectroscopy (XPS) data in Fig. 1a, Supplementary Figs. 4–7 and Supplementary Table 1 confirm the uniform spatial distribution of all constituent elements, namely, I, Pb, Rb, C, N and O, within the single crystal. This homogeneity provides strong evidence that the synthesized compound has a well-defined stoichiometry and that the crystal growth process successfully incorporated all starting elements into a single, coherent phase. The absence of elemental segregation or secondary phases supports the high quality of the grown crystal.

In addition, we envision that Rb^+ and MA^+ may co-occupy the A-site in the 1D perovskite framework when MA^+ is present. To verify this, we recrystallized single crystals in the presence of MACl. EDS mapping and XPS (Supplementary Figs. 8–10 and Supplementary Table 2) confirm a homogeneous distribution of all relevant elements (I, Pb, Cl, Rb, C, N and O) throughout the crystal. Quantitatively, the measured Rb content reduces when MA^+ is included, providing strong evidence for A-site co-occupancy by MA^+ and Rb^+ in the resulting perovskite lattice.

The corresponding single-crystal structures were resolved and refined using SHELXT and OLEX2^{24–26}. The compound crystallizes in the orthorhombic crystal system with space group *Ibam* (no. 72), a centrosymmetric space group commonly associated with high symmetry. Details of X-ray crystallographic parameters of the single crystals are provided in Supplementary Tables 3 and 4.

Single-crystal X-ray diffraction analysis, complemented by structural simulations, reveals a structural modification of the common lead-iodide octahedral motif (Supplementary Fig. 2). In the classic $[PbI_6]^{4-}$ octahedron, two of the six iodide (I^-) ligands are replaced by two OCN^- anions. This substitution is noteworthy because the OCN^- ion is a pseudo-halide with a linear $[O-C\equiv N]^-$ geometry, yet it coordinates to the Pb^{2+} centre through its N atom, forming a bond between Pb^{2+} and $[O-C\equiv N]^-$. The resulting coordination unit can be described as a distorted $[PbI_4(OCN)_2]^{4-}$ octahedron. Furthermore, these modified octahedra share edges with each other, forming dimeric or potentially extended 1D chains. This edge-sharing connectivity is a crucial structural feature that distinguishes the material from corner-sharing perovskite-like networks and probably influences its electronic structure and stability (Fig. 1b).

A key finding from the structural refinement is the disordered nature of the OCN^- ions within the crystal lattice. The analysis suggests that the OCN^- ions are not statically fixed in a single orientation but rather exist in a state of rotational or orientational disorder, showing at least three distinct spatial arrangements, as shown in Fig. 1b. This type of disorder is frequently attributed to thermal motion at the molecular level. The relatively low energy barrier for rotation of the linear OCN^- ion, combined with the spacious voids and flexible coordination environment provided by the edge-sharing dimer structure, allows these anions to dynamically sample multiple equivalent positions within the lattice (Supplementary Fig. 11). This dynamic disorder is a common feature in hybrid or inorganic crystals containing linear anions or molecules and can have profound implications for the material's physical properties, such as dielectric response, ionic conductivity and phase transition behaviour²⁷. Inspired by the similar structures and properties of these cyanates and pseudohalides, we were also able to synthesize a series of comparable single crystals, namely, $MAPbI_2OCN$, $KPbI_2OCN$, $CsPbI_2OCN$, $RbPbI_2SCN$ and $MAPbI_2SCN$, all of which show structures similar to that of $RbPbI_2OCN$ (Supplementary Fig. 3).

To rationalize the optoelectronic properties of $RbPbI_2OCN$, we examined its electronic structure via density functional theory calculations. The unfolded band structure (Fig. 1c) shows an indirect bandgap of ~ 1.24 eV, a value that falls within the Shockley–Queisser optimal range for single-junction photovoltaics. Crucially, the electronic structure deviates fundamentally from archetypal lead halide perovskites owing to the synergistic coupling between the inorganic Pb–I framework and the dynamically disordered OCN^- anion²⁸.

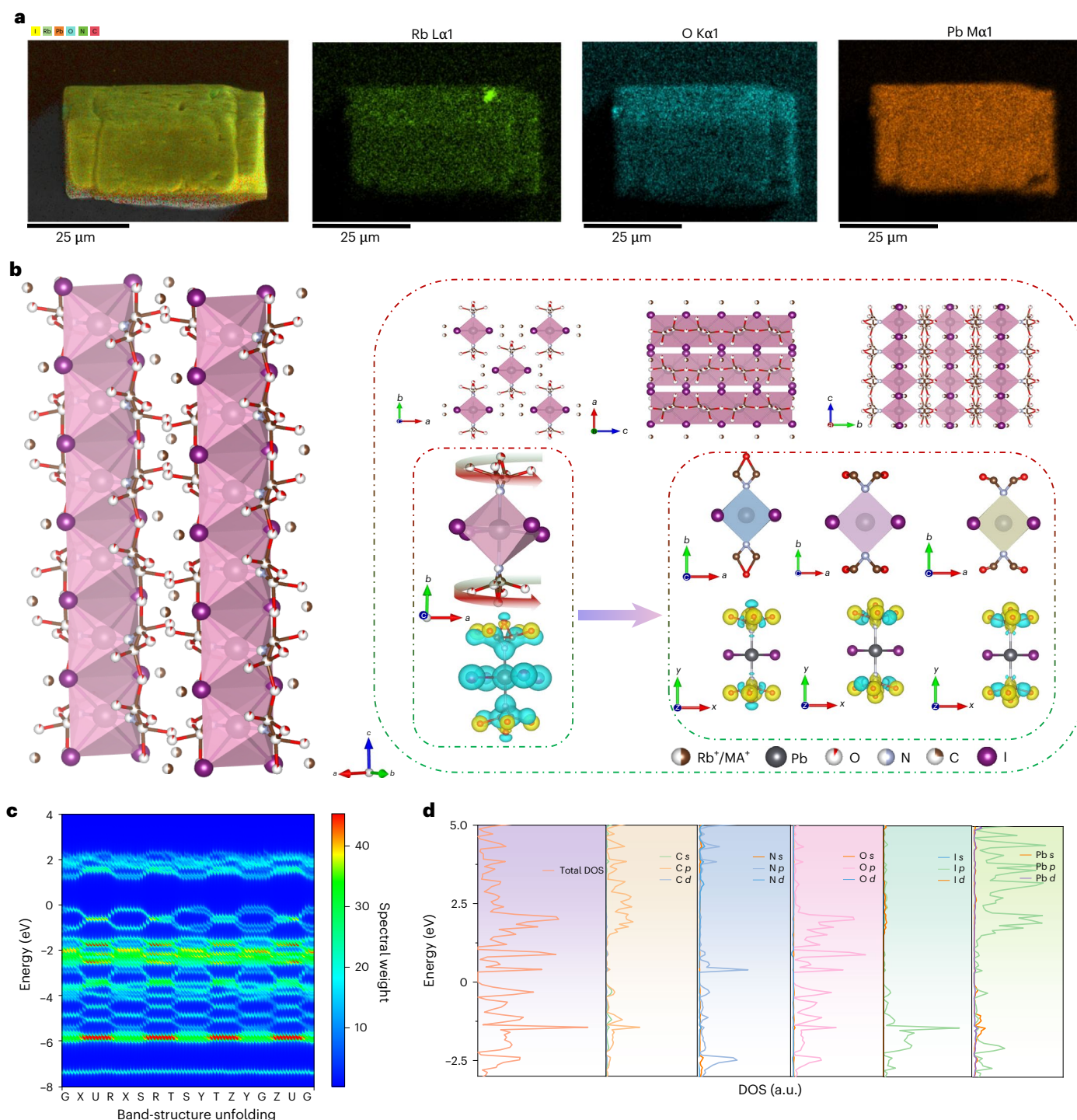


Fig. 1 | Structural and electronic characteristics of single-crystal RbPbI₂OCN.

a, EDS mapping images of RbPbI₂OCN showing the uniform spatial distribution of I, Pb, Rb, C, N and O within the single crystal (Supplementary Fig. 4).

b, Dynamically disordered mode of RbPbI₂OCN and corresponding differential charges. **c**, Band structure of RbPbI₂OCN calculated by density functional theory. **d**, Projected DOS of the different elements in RbPbI₂OCN.

As shown in the projected density of states (DOS) (Fig. 1d), while the valence band maximum remains primarily defined by Pb 6s–15p antibonding interactions, the OCN[−] anion introduces distinct states in the deep valence region through its O 2p and C/N 2p orbitals. The conduction band minimum, dominated by Pb 6p character, is also subtly modified by weak hybridization with OCN[−] π* orbitals, leading to reduced band dispersion and an increased electron effective mass. This unique role of the OCN[−] anion as an active electronic modulator, rather than a passive structural component, offers an unconventional

pathway to tailor the optoelectronic properties of hybrid perovskites beyond traditional compositional engineering.

Crystallization kinetics of bulk heterojunctions

The in situ photoluminescence (PL) images of the crystallization nucleation process of perovskite films with different RbOCN concentrations after the addition of the anti-solvent ethyl acetate are shown in Fig. 2a. By integrating the single-crystal structural characterization of RbPbI₂OCN with in situ crystallization kinetic studies, a coherent and multiscale

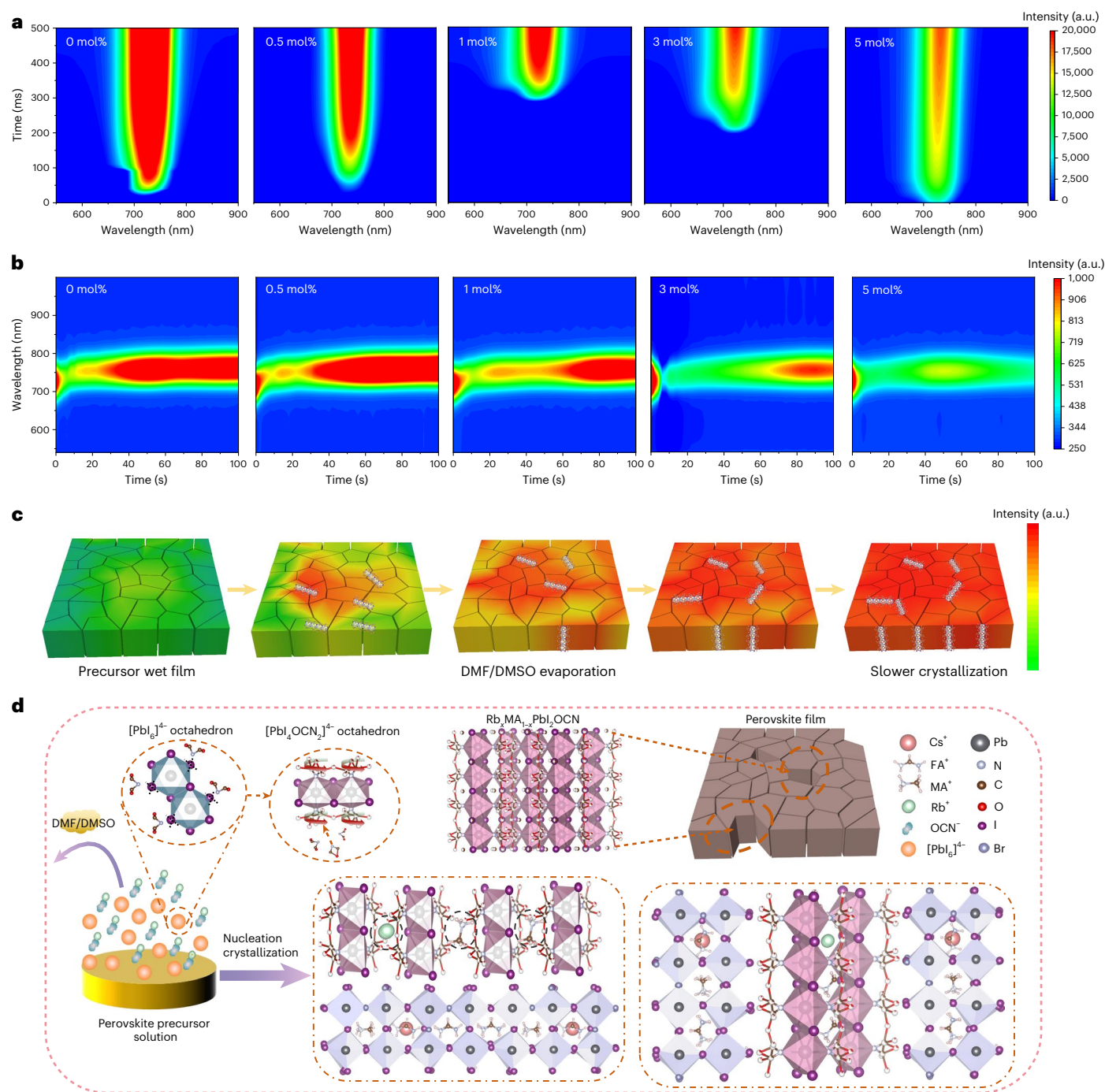


Fig. 2 | Crystallization kinetics for the formation of perovskite bulk heterojunctions. **a**, In situ PL images of crystallization nucleation process of perovskite films with varying contents of RbOCN and MACl during anti-solvent ethyl acetate treatment. **b**, In situ PL images of the corresponding perovskite films illustrating the nucleation and growth process during annealing at 80 °C.

c, Schematic diagram of nucleation, crystal growth and the formation of perovskite bulk heterojunctions. The colour bar represents the intensity of the crystallization process, ranging from green (low) to red (high). **d**, Possible locations of RbPbI₂OCN in perovskite bulk heterojunctions.

mechanism underlying the role of RbOCN is established. Central to this mechanism is the distinct structural and electronic nature of the OCN⁻ anion, which actively participates in and directs the dynamic transformation from the precursor solution to the final solid-state film. The observed prolongation of nucleation time upon RbOCN addition is a classic signature of a coordination-mediated nucleation inhibition mechanism. In the precursor solution, Pb²⁺ ions are typically coordinated by DMSO, formamidinium (FA⁺) and I⁻. The introduced OCN⁻ ion, with its oxygen donor atom, has a strong affinity for Pb²⁺, competing

with these solvents, as revealed by in situ X-ray diffraction analysis (Supplementary Fig. 12). A peak corresponding to the PbI₂-DMSO complex was observed at low temperature, which gradually disappeared and transformed into the pure PbI₂ phase without the formation of any perovskite-related phases. In contrast, in the presence of OCN⁻, PbI₂ first coordinates with DMSO at the low-temperature stage to form the PbI₂-DMSO complex. Upon heating, OCN⁻ progressively replaces DMSO in the Pb²⁺ coordination sphere, leading directly to the crystallization of the 1D RbPbI₂OCN perovskite phase. This ligand-exchange-driven

process lowers the crystallization activation energy, enabling controlled grain growth with reduced defect density.

Again, compared with the control sample, the RbPbI_2OCN -containing precursor shows a prolonged crystallization time, indicating slower and more controlled nucleation (Fig. 2a,b). This slower nucleation reduces the initial number of crystal nuclei, providing each growing grain with more space to develop without competitive impingement²⁹. At the optimal concentration of 1 mol%, this moderation achieves a balance by preventing explosive nucleation, which leads to small grains and pinholes. Figure 2c and Supplementary Fig. 13 show the schematic diagram and the film change diagram of nucleation, crystal growth and the formation of perovskite bulk heterojunctions. As illustrated in Fig. 2d, the edge-sharing $[\text{PbI}_4(\text{OCN})]^{4-}$ dimer structure of RbPbI_2OCN is structurally distinct from the corner-sharing $[\text{PbI}_6]^{4-}$ network of the α -phase perovskite. This heterostructure cannot be epitaxially integrated into the perovskite lattice without strain. Therefore, these nanodomains or monolayers of RbPbI_2OCN act as passivating caps. They impede the rapid fusion of adjacent perovskite grains, forcing growth to proceed more slowly through ion diffusion around these barriers, ultimately leading to larger, more highly oriented grains with fewer defects^{30–33}.

To verify this hypothesis, we conducted time-of-flight secondary ion mass spectrometry depth profiling and elemental mapping (Fig. 3a and Supplementary Figs. 14 and 15), which further reveal that the 1D phase is predominantly localized at grain boundaries, forming a homogeneous distribution around the 3D perovskite grains. Likewise, scanning electron microscopy (SEM) images further demonstrate that the incorporation of RbOCN induces preferential orientation growth of the perovskite crystals, improves crystallization quality and suppresses grain-boundary density (Supplementary Fig. 16). However, an excess of RbOCN disrupts the crystal structure and introduces impurity phases, leading to deteriorated crystallinity. The target perovskite film treated with 1 mol% RbOCN shows a pronounced increase in grain size and the presence of uniformly distributed RbPbI_2OCN within the perovskite bulk films, with grains showing a preferential vertical orientation relative to the substrate (Fig. 3b,c and Supplementary Figs. 17 and 18). This microstructural evolution is a direct consequence of the modified crystallization kinetics induced by the in situ-formed RbPbI_2OCN phase. The 1D perovskite achieves efficient elimination of PbI_2 defects in the perovskite film by coordinating with free PbI_2 , chemically passivating defect sites and regulating energy levels. In addition, the hydrophobicity of the surface of the perovskite film is increased (Supplementary Figs. 19 and 20). This continuous and dense microstructure, with reduced grain-boundary density, establishes a direct, low-tortuosity pathway for charge carriers. The vertical alignment of grains facilitates efficient charge extraction towards the electrodes, reducing bulk recombination and series resistance, which is foundational for achieving a high fill factor (FF) and current density (J_{sc}).

To probe the interface within this engineered microstructure, we performed EDS elemental mapping and high-resolution transmission electron microscopy (HRTEM) on the target film. The EDS elemental mapping and line-scan analysis of the perovskite thin films further verify the presence of 1D phase components, which are homogeneously distributed along the grain boundaries of the 3D perovskite host, thereby forming a continuous 1D/3D heterojunction structure (Supplementary Fig. 21 and Supplementary Table 5). Figure 3d shows an HRTEM image of the heterojunction region, clearly revealing the lattice fringes at the interface between the 3D perovskite matrix and the 1D RbPbI_2OCN phase. The measured lattice spacings from the HRTEM images are in excellent agreement with the calculated interplanar distances derived from the single-crystal X-ray diffraction structure. The observed lattice coherence suggests favourable interfacial matching, which is critical for minimizing interfacial defect states. Theoretical calculations support this, indicating that the distinct edge-sharing structure of the 1D phase can form a low-strain

heterojunction with the corner-sharing 3D network. This structurally coherent interface, combined with the strong Pb–N coordination in the 1D phase, provides robust chemical passivation of under-coordinated Pb^{2+} and I[−] vacancies at grain boundaries, which is a primary source of non-radiative recombination.

Differential charge density analysis was performed on a crystallographically constructed RbPbI_2OCN /perovskite heterojunction model. As shown in Fig. 3e, the two phases show good lattice matching through shared $[\text{PbI}_6]^{4-}$ octahedra and intermolecular interactions. The charge accumulation (yellow) and depletion (cyan) regions reveal interfacial charge redistribution, with electron transfer from the RbPbI_2OCN phase to the perovskite layer. Importantly, the continuous charge redistribution across the interface indicates strong electronic coupling beyond weak van der Waals interactions, originating from orbital hybridization between the two components. These results directly demonstrate the electronic interactions and charge-separation behaviour at the heterojunction, supporting the proposed electronic structure and interfacial properties.

In addition, density functional theory calculations reveal that OCN^- adsorbs strongly on the perovskite surface with an energy of -5.349 eV, indicating a robust chemical interaction that can modify the surface energy and influence crystallization kinetics. Furthermore, DOS analysis shows that the pristine perovskite surface shows pronounced defect-induced gap states arising from under-coordinated Pb^{2+} ions and iodine vacancies (Supplementary Fig. 22). Upon OCN^- adsorption, these gap states are effectively suppressed and nearly eliminated, providing direct quantitative evidence that the strong Pb^{2+} – OCN^- interaction effectively passivates surface defects, reduces non-radiative recombination centres and restores a bulk-like electronic structure.

To gain deeper insight into the charge-transfer and recombination mechanisms, we performed PL and time-resolved PL (TRPL) measurements. Compared with the control film, the target film shows enhanced PL intensity (Supplementary Fig. 23) and a prolonged carrier lifetime (Fig. 3f and Supplementary Table 6), indicating a reduction in defect density and non-radiative recombination³⁴. This conclusion is further supported by space-charge-limited current measurements, which revealed lower trap densities ($1.55 \times 10^{16} \text{ cm}^{-3}$ and $6.42 \times 10^{15} \text{ cm}^{-3}$) and higher charge carrier mobilities ($7.73 \times 10^{-2} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $1.69 \times 10^{-1} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) in the target hole- and electron-only devices, respectively, compared with the control values of $2.01 \times 10^{16} \text{ cm}^{-3}$ and $7.07 \times 10^{15} \text{ cm}^{-3}$, and $5.30 \times 10^{-2} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $1.11 \times 10^{-1} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ (Supplementary Fig. 24).

Single-junction wide-bandgap perovskite solar cells

Single-junction wide-bandgap perovskite solar cells were fabricated with the following structure: tin-doped indium oxide (ITO)/ NiO_x /[4-(3,6-dimethoxy-9H-carbazol-9-yl)butyl]phosphate (MeO-4PACz)/perovskite/ethylenetriamine dihydroiodide (EDAI_2)/ $\text{C}_{60}/\text{SnO}_2/\text{Ag}$. We first compared the photovoltaic parameters of devices incorporating a series of pseudo-halide and alkali-cyanate-based 1D perovskite derivatives (Supplementary Fig. 25). These 1D phase-incorporated devices showed improved performance compared with the control devices, indicating that this family of 1D perovskite-derived materials is broadly applicable and offers a versatile platform for optimizing device performance. Among these, devices incorporating RbPbI_2OCN showed superior performance and were selected as target devices for further characterization and photovoltaic parameter optimization. Notably, although solar cells incorporating either Rb^+ or OCN^- alone showed improved performance, the 1D RbPbI_2OCN perovskite derivatives featuring dynamically disordered cyanate anions showed a synergistic role of Rb^+ and OCN^- in this scenario (Supplementary Fig. 25), resulting in improved photovoltaic performance compared with systems containing only Rb^+ or OCN^- . The target device shows improved energy-level alignment with charge-transport layers (Supplementary Figs. 26 and 27).

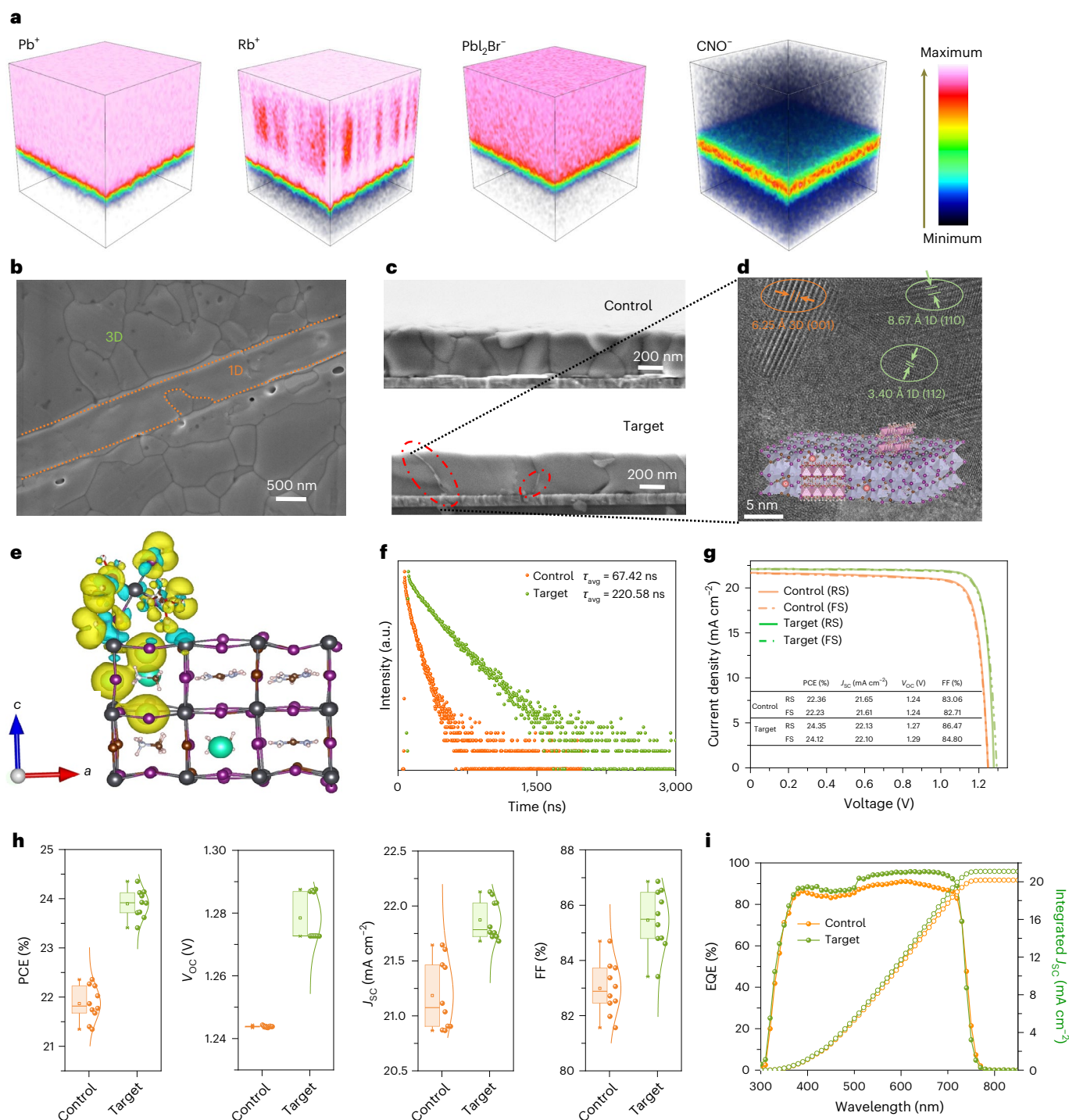


Fig. 3 | Device configuration and photovoltaic performance of single-junction perovskite solar cells. a, Time-of-flight secondary ion mass spectrometry 3D tomography images (ion signal intensity from black to pink) of the 1D/3D heterojunction perovskite film. **b**, SEM image of the 1D/3D heterojunction perovskite film. **c**, Cross-sectional SEM images of the control and target films. **d**, HRTEM image of the perovskite bulk heterojunctions. **e**, Differential charge density maps of the RbPbI₂OCN/perovskite heterojunction interface. The yellow and cyan regions represent charge accumulation and charge depletion, respectively. **f**, TRPL spectra of the control and target perovskite films. **g**, J - V curves for the control and target

devices. RS and FS denote reverse scan and forward scan, respectively, during the J - V measurement. The solid lines represent the reverse scan, while the dashed lines represent the forward scan. **h**, Photovoltaic parameter statistics (PCE, V_{oc} , J_{sc} and FF). The data are presented as the median (solid line within the box) and mean (hollow square), with 10 devices (spheres) under each group condition; the lower and upper box lines represent the 25–75% quartiles, and the whiskers denote the 1.5 $\times$ interquartile range. **i**, EQE spectra of the control and target devices. The solid circles represent the EQE spectra, while the open circles represent the integrated J_{sc} values derived from the EQE measurements.

This enhancement is probably attributed to the electronic contribution of the OCN⁻ anions in the 1D phase, thus improving the local DOS and work function at interfaces and grain boundaries. After optimizing the composition of the RbPbI₂OCN perovskite derivative, specifically

the contents of RbOCN and MACl (Supplementary Figs. 28 and 29 and Supplementary Table 7), the target devices showed substantial enhancements in J_{sc} , V_{oc} and FF compared with the control device, ultimately delivering both a high PCE and excellent reproducibility

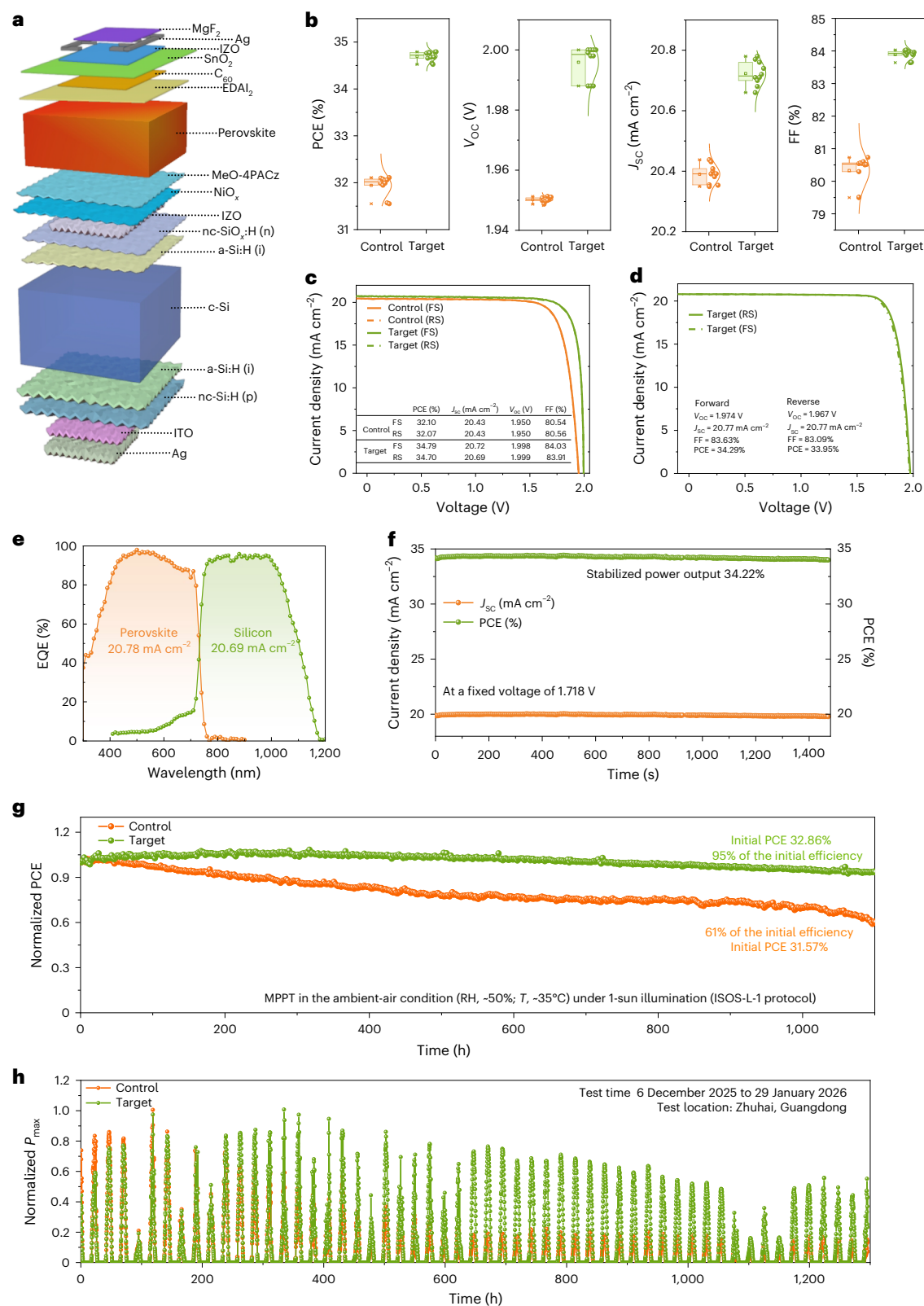


Fig. 4 | Device performance of TSCs. **a**, Schematic of two-terminal perovskite-crystalline silicon (c-Si) TSCs. nc-Si:H (p), hydrogenated phosphorus-doped nanocrystalline silicon (p type); nc-SiO_x:H (n), hydrogenated boron-doped nanocrystalline silicon oxide (n type); a-Si:H (i), hydrogenated amorphous silicon (intrinsic). **b**, Photovoltaic parameter statistics of perovskite-silicon TSCs (PCE, J_{SC} , V_{OC} and FF). The data are presented as the median (solid line within the box) and mean (hollow square), with 10 devices (spheres) under each group condition; the lower and upper box lines represent the 25%–75% quartiles, and the whiskers

denote the 1.5 $\times$ interquartile range. **c**, J - V curves of TSCs in the laboratory. **d**, J - V curves of TSCs. **e**, EQE spectra of the target TSC. **f**, Maximum power output of the TSC under 1-sun illumination. **g**, Maximum power point tracking (MPPT) of the encapsulated control and target TSCs measured under continuous 1-sun illumination. RH denotes the relative humidity of the testing atmosphere. **h**, Outdoor stability test of the encapsulated TSCs under the ISOS-O-2 protocol.

(Fig. 3h). The optimized target device achieved a PCE of 24.35% with a J_{sc} of 22.13 mA cm⁻², a V_{oc} of 1.27 V and a FF of 86.47% (Fig. 3g).

The external quantum efficiency (EQE) spectra yielded integrated J_{sc} values of 20.27 mA cm⁻² and 21.21 mA cm⁻² for the control and target devices, respectively, highlighting enhanced photon absorption, optimized optical bandgap and tailored electronic structure capability of the RbPbI₂OCN-incorporating device across the full spectral range (Figs. 1b–d and 3i and Supplementary Fig. 30). Such improvement can be further attributed to enhanced charge transport within vertically aligned large grains and reduced energy losses in the bulk phases and/or at interfaces¹⁷. Moreover, unencapsulated target devices show enhanced thermal stability compared with the control device, retaining 82.2% of their initial efficiency after 900 h of operation at 85 °C (International Summit on Organic Photovoltaic Stability (ISOS)-D-2I protocol) (Supplementary Fig. 31).

Improved bulk and interfacial properties were also evidenced by transient photovoltage decay and dark current density–voltage (J – V) measurements, with the target device showing an extended carrier lifetime (Supplementary Fig. 32) and reduced leakage current (Supplementary Fig. 33). The dependence of V_{oc} on incident light intensity further corroborates these findings, showing a lower ideality factor for the target device ($1.59 k_B T/q$) than the control device ($1.70 k_B T/q$) (Supplementary Fig. 34) (where k_B is the Boltzmann constant, T is the absolute temperature, and q is the elementary charge). In addition, the calculated V_{oc} loss in the target device was below 0.4 V, substantially lower than that of the control one (Supplementary Table 8). These improved properties, along with suppressed non-radiative recombination, account for the superior V_{oc} and FF values observed in target devices. Collectively, the results further confirm that incorporating the RbPbI₂OCN perovskite derivative promotes a shallow-acceptor-dominated, defect-tolerant electronic environment, thereby facilitating enhanced carrier extraction and transport (Fig. 1c).

Monolithic perovskite–silicon TSCs

We then fabricated monolithic two-terminal perovskite–silicon TSCs, using a silicon heterojunction solar cell as the bottom subcell, whose photovoltaic performance is shown in Supplementary Fig. 35. The top cell adopted the following architecture: indium zinc oxide (IZO)/NiO_x/MeO-4PACz/perovskite/EDAI₂/C₆₀/SnO₂/IZO/Ag/MgF₂, as schematically shown in Fig. 4a. Statistical boxplot analysis was performed on tandem cells on the basis of both pristine 3D wide-bandgap perovskites and engineered coherent bulk heterojunctions incorporating RbPbI₂OCN in the top cells (Fig. 4b). The results reveal that tandem devices incorporating the bulk heterojunction show excellent reproducibility and achieve an impressive PCE of 34.79%, with a V_{oc} of 1.998 V, an FF of 84.03% and a J_{sc} of 20.72 mA cm⁻² (Fig. 4c and Supplementary Table 9), outperforming the control devices. Meanwhile, the target device showed a steady-state PCE of 34.22% at a fixed voltage of 1.718 V after continuous maximum power point tracking operation for 1,400 s (Fig. 4f). This overall enhancement further confirms the functional role of the RbPbI₂OCN phase within the top cell. One of the optimized target devices was sent to the China Photovoltaic Testing Center (CPVT) for independent certification, obtaining a certified PCE of 34.29% with a V_{oc} of 1.974 V, J_{sc} of 20.77 mA cm⁻² and FF of 83.63% (Fig. 4d and Supplementary Fig. 36). The EQE spectra of the target tandem device is presented in Fig. 4e. The integrated J_{sc} values are calculated to be 20.78 mA cm⁻² and 20.69 mA cm⁻² for the perovskite top cell and silicon bottom cell, respectively, indicating excellent current matching between the two subcells. Again, the precise current matching and exceptional tandem performance are ultimately enabled by the structurally coherent and electronically passivated heterojunction templated by the unique 1D RbPbI₂OCN phase, demonstrating how rational anionic engineering can translate microscopic structural control into macroscopic photovoltaic excellence.

Operational stability was evaluated by performing maximum power point tracking on encapsulated control and target devices under continuous 1-sun illumination in ambient air (humidity, ~50%; temperature, ~35 °C) under the ISOS-L-1 protocol. As shown in Fig. 4g, the control device retained only about 61% of its initial PCE after 1,100 h, while the target device maintained 95% of its initial efficiency over the same period. We also measured the thermal stability of both tandem devices during operation at 85 °C (ISOS-D-2I protocol). The control device retained only ~80% of its initial PCE after 500 h of operation, whereas the target device retained over 90% of its initial PCE after 1,000 h (Supplementary Fig. 37). Extended real-world testing over 1,300 h, conducted in accordance with the ISOS-O-2 protocol³⁵ further confirmed the enhanced stability of the target tandem device (Fig. 4h). PL spectra for both the pristine perovskite film and its corresponding tandem device under ageing conditions revealed obvious phase segregation of wide-bandgap perovskites, whereas the target samples maintained a stable perovskite phase (Supplementary Fig. 38). This enhanced stability is attributed to the multifunctional role of the 1D RbPbI₂OCN phase located at grain boundaries or interfaces, which provides robust chemical passivation and acts as a phase stabilizer, thereby reducing the density of degradation-prone defect sites. Overall, the exceptional operational stability is a direct consequence of the rationally engineered 1D RbPbI₂OCN phase. Its unique edge-sharing structure and cyanate-based chemistry not only provide robust defect passivation but also template a compact, large-grained bulk-heterojunction architecture. We thereby establish a transformative paradigm in which atomic-scale anionic design dictates mesoscale film morphology and macroscopic device stability, bridging materials innovation with photovoltaic performance at an unprecedented level.

Outlook

We establish a materials design strategy for perovskite photovoltaics by rational engineering of anionic chemistry. By designing and synthesizing a series of low-dimensional perovskite derivatives, namely, MAPbI₂OCN, KPbI₂OCN, CsPbI₂OCN, RbPbI₂SCN and MAPbI₂SCN, we introduce a dynamically disordered anionic structure that fundamentally transforms crystallization dynamics, interfacial properties and defect physics. The OCN⁻/SCN⁻ anions function as multiscale structural directors, enabling the orientational growth of coherent bulk heterojunctions while providing atomic-level chemical passivation of grain boundaries and surfaces. This dual functionality enables unprecedented control over the morphology–electronic property relationship in wide-bandgap perovskites. The resultant single-junction solar cell achieves a competitive efficiency of 24.35% for its bandgap class, while the monolithic perovskite–silicon tandem architecture reaches an impressive efficiency of 34.79% (certified 34.29%). The tandem device also shows exceptional operational stability, maintaining 95% of its initial performance after 1,100 h under the ISOS-L-1 protocol and enhanced outdoor stability over 1,300 h under the ISOS-O-2 protocol. We demonstrate a generalizable strategy for engineering the anionic sublattice with dynamically disordered molecular ions, such as OCN⁻ and SCN⁻. This approach reconstitutes the perovskite lattice, creating an intrinsically stable and high-performing material genotype.

Methods

Materials

Formamidinium iodide (FAI, ≥99.5%) was purchased from Great-cell Solar. Methylammonium bromide (MABr, ≥99.995%), MAI (≥99.995%), PbI₂ (≥99.99%), lead bromide (PbBr₂, 99.99%), caesium iodide (CsI, 99.999%), rubidium cyanate (RbOCN), methylammonium cyanate (MAOCN), rubidium thiocyanate (RbSCN), caesium cyanate (CsOCN), potassium cyanate (KOCN), methylammonium thiocyanate (MASCN), rubidium iodide (RbI), ethylenetriamine dihydroiodide (EDAI₂, ≥99.0%) and [4-(3,6-dimethoxy-9H-carbazol-9-yl)butyl] phosphate (MeO-4PACz, ≥99.0%) were purchased from Xi'an e-Light.

Phenethylammonium (>99.5%) and C₆₀ were purchased from Xi'an Polymer Light Technology. DMSO (≥99.9%), *N,N*-dimethylformamide (DMF, 99.8%), isopropanol (IPA, ≥99.9%) and ethyl acetate (EA, ≥99%) were purchased from Aladdin. All materials were used as received without further purification.

Synthesis of single crystals

PbI₂ was individually mixed with RbOCN, MAOCN, KOCN, CsOCN, RbSCN and MASCN in DMSO solution at a molar ratio of approximately 5:1 (for example, 542 mg PbI₂ with 30 mg RbOCN in 500 μl DMSO). Excess PbI₂ was used to obtain a saturated solution. Each mixture was shaken using a vortex mixer at room temperature for 1 h and then filtered to obtain a transparent precursor solution. The solution was transferred to a 2 ml glass vial, which was placed uncapped inside a 20 ml glass vial. Next, 1 ml of ethyl acetate solvent was added to the larger vial to serve as a counter-solvent for vapour-assisted crystallization. The large vial was sealed and kept at ambient temperature for 2 days, during which white needle-like single crystals grew.

Computational details

The theoretical calculations were performed using Real Space Electronic Structure Calculator, Nanodcal and DS-PAW software in Device Studio of Hongzhiwei Technology (Shanghai)^{36–38}.

Here the electronic structure of a 1D single crystal is calculated using the first-principles method. The following electronic structure calculations were performed using the density functional theory expansion of the Kohn–Sham equation:

$$\left[-\frac{1}{2}\nabla^2 + \int d\mathbf{r}' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} + V_{\text{ion-e}}(\mathbf{r}) + V_{\text{ext}}(\mathbf{r}) + V_{\text{xc}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r}) \quad (1)$$

where ∇^2 is the Laplace operator, $\rho(\mathbf{r}')$ is the electron density distribution, $V_{\text{ion-e}}$ is the interaction energy between electrons and ions or nuclei, $V_{\text{ext}}(\mathbf{r})$ is the potential energy of the external electric or magnetic field, $\psi_i(\mathbf{r})$ is the wave function of the i th Kohn–Sham orbital and ε_i is the corresponding energy eigenvalue. $V_{\text{xc}}(\mathbf{r})$ is the exchange–correlation potential, describing the quantum exchange between electrons acting on the correlation effect, which is usually calculated by an approximate method:

$$V_{\text{xc}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}}{\delta \rho(\mathbf{r})} \quad (2)$$

where E_{xc} is the exchange–correlation energy. The exchange–correlation function is defined by the generalized gradient approximation of Perdew–Burke–Ernzerhof and calculated by a linear combination of atomic orbitals.

The band structure and DOS calculations were performed using DS-PAW software. In the Brillouin region of the system, the Monkhorst–Pack grid with $9 \times 8 \times 7$ density was selected for sampling. The real spatial resolution was set to 0.25 Bohr, and the convergence accuracies for energy and charge density were higher than 1×10^{-6} Hartree and $1 \times 10^{-6} e$. The calculations were performed under the condition of spin degeneracy.

DOS calculation description

The DOS calculations were performed on the basis of the converged self-consistent field (SCF) electronic structure using the Nanodcal code integrated within the Hongzhiwei Device Studio platform. The local density approximation with the Perdew–Zunger (PZ81) exchange–correlation functional was adopted, combined with the double-zeta polarized numerical atomic orbital basis set. A real-space cut-off energy of 80 Hartree and a $3 \times 3 \times 1$ Monkhorst–Pack k -point mesh were used for SCF relaxation. For the DOS calculations, a denser $3 \times 3 \times 3$ k -point grid was used to sample the Brillouin zone, with an energy range from

–8.0 eV to 8.0 eV and 201 energy sampling points to ensure smooth, accurate DOS curves. Orbital-projected DOS calculations were performed to decompose the electronic states into contributions from individual atomic orbitals, enabling a detailed analysis of the electronic structure and its correlation with the transport properties of the chiral perovskite system. All calculations were implemented using the standardized workflow in Device Studio, ensuring the reproducibility of the parameter settings and results.

Description of differential charge calculation and processing

The defect DOS calculations were performed on the basis of the converged SCF electronic structure using the Nanodcal code integrated within the Hongzhiwei Device Studio platform. The calculations used the local density approximation with the Perdew–Zunger exchange–correlation functional (PZ81) and a double-zeta polarized numerical atomic orbital basis set. A real-space cut-off energy of 80 Hartree was adopted, and a $6 \times 6 \times 2$ Monkhorst–Pack k -point mesh was used to sample the Brillouin zone during SCF relaxation, ensuring sufficient convergence of the electronic structure for the defect-containing system.

For the defect DOS analysis, an energy range from –10.0 eV to 10.0 eV with 401 energy sampling points was selected to accurately resolve the defect energy levels near the Fermi level and the state density distribution at the band edges. No orbital projection was applied (plot = 'None'), focusing on the global electronic structure characterization. This study aims to reveal the regulatory role of intrinsic defects in the electronic states of chiral perovskites, providing theoretical support for the optimization of device transport properties.

Adsorption energy calculation description

The total energy calculation was performed using a converged self-consistent-field electronic structure using the Nanodcal code, integrated within the Hongzhiwei Device Studio platform. The calculation used the converged NanodcalObject.mat as input to directly compute the total energy of the system, providing fundamental data for subsequent thermodynamic property analyses such as binding energy and stability.

Preparation of perovskite precursor solution

To prepare the pristine perovskite solution, 18.8 mg CsI, 25.9 mg MABr, 198.8 mg FAI, 178.7 mg PbBr₂ and 599.9 mg PbI₂ were dissolved in 1 ml of a DMF/DMSO mixed solvent (4:1 v/v). Separately, 5 mg RbOCN and 5 mg MACl were dissolved in 1 ml DMSO to prepare the additive solution. Both solutions were heated at 60 °C under continuous stirring until complete dissolution. Each fully dissolved solution was then filtered individually. Subsequently, the filtered additive solution was mixed with the pristine perovskite solution at a volume ratio of 1:10.

Preparation of single-junction wide-bandgap perovskite cells

ITO-coated glass substrates were sequentially cleaned with detergent, deionized water and ethanol by ultrasonication (20 min per step) and then dried in an oven. After a 20 min pre-sputtering step, a 20-nm-thick NiO_x layer was prepared onto the ITO substrate by sputtering a NiO_x target (99.9%) at room temperature using a radio-frequency magnetron sputtering deposition system with an argon gas flow rate of 20 sccm, a power of 65 W and a working pressure of 2 Pa for 20 min, followed by post-annealing in air at 200 °C for 20 min (ref. 39). A self-assembled monolayer (MeO-4PACz) was subsequently formed on the NiO_x surface by spin-coating a solution of MeO-4PACz in isopropanol (0.5 mg ml^{–1}) at 3,000 rpm for 30 s, followed by annealing at 100 °C for 10 min. To remove excess unanchored MeO-4PACz, isopropanol was spin-coated on the ITO/NiO_x/MeO-4PACz substrate at 3,500 rpm for 30 s. The substrates were then transferred into a dry-air glovebox. The perovskite solution was deposited on the ITO/NiO_x/MeO-4PACz substrates using a sequential two-step procedure: 1,000 rpm for 10 s, followed by 6,000 rpm for 30 s. In the second step, 240 μl ethyl acetate was

dropped onto the spinning substrate 10 s before the end of the programme. After annealing at 100 °C for 20 min, an EDAl_2 solution in isopropanol (0.5 mg ml^{-1}) was spin-coated onto the perovskite film at 4,000 rpm for 30 s and annealed at 100 °C for 10 min. Subsequently, a 30-nm C_{60} layer was thermally evaporated (QHV-R53). An ALD- SnO_2 layer (~30 nm) was then prepared by an atomic layer deposition (ALD) system (Dongguan Nanofrontier Microelectronics) using TDMASn and H_2O as precursors at 100 °C for 200 cycles. Finally, 800-nm Ag top electrodes were deposited by a thermal evaporator (QHV-R53). The devices have an overall dimension of $2.5 \text{ cm} \times 2.5 \text{ cm}$ and an effective illuminated area of 0.09 cm^2 .

Fabrication of two-terminal perovskite–silicon TSCs

The silicon bottom cells were fabricated on M6 float-zone n-type wafers (280 μm thickness, 1–3 $\Omega \text{ cm}$) using a commercial silicon heterojunction (SHJ) research and development line. These wafers served as substrates for the subsequently fabricated perovskite–silicon TSCs with micro-textured surfaces. The wafers were first cleaned with ozone and then immersed in a potassium hydroxide solution containing additives to form double-sided submicrometre pyramids. Next, the wafers underwent wet-chemical treatments, including chemical pyramid rounding and Radio Corporation of America cleaning.

Following the wet-chemical processes, the wafers were transferred to a plasma-enhanced chemical vapour deposition (PECVD) system for silicon-based thin-film deposition. A radio-frequency PECVD system (13.56 MHz) was first used to deposit an ~5-nm hydrogenated intrinsic a-Si:H layer on both sides of the wafer using silane (SiH_4) and hydrogen (H_2) as precursor gases. The deposition was carried out at a pressure of 60 Pa and a temperature of 210 °C. Subsequently, very-high-frequency PECVD at 40 MHz was used to deposit the doped contact layers. On the front side, a 15-nm hydrogenated phosphorus-doped nanocrystalline silicon oxide (n-nc- $\text{SiO}_x\text{:H}$) layer was formed to selectively extract electrons. The deposition was performed at a pressure of 310 Pa and a temperature of 210 °C, with a high hydrogen dilution ratio to promote crystallization. n-type doping and oxygen incorporation were achieved by introducing phosphane (PH_3) and carbon dioxide (CO_2) into the $\text{SiH}_4\text{:H}_2$ plasma. On the rear side, a CO_2 plasma treatment was first performed on the intrinsic a-Si:H layer to form a thin barrier layer that prevents plasma damage. Then, the rear hydrogenated boron-doped nanocrystalline silicon (p-nc-Si:H) emitter (~20 nm) was prepared by introducing diborane boroethane (B_2H_6) into the $\text{SiH}_4\text{:H}_2$ plasma using very-high-frequency PECVD. This deposition was carried out at a pressure of 300 Pa and a temperature of 210 °C, also with a high hydrogen dilution ratio to promote crystallization. To complete SHJ bottom-cell fabrication, transparent conductive oxide layers were deposited by direct-current magnetron sputtering at room temperature using an argon–oxygen mixture as the reaction gas. On the rear side, a 70-nm ITO layer (50 ohms per square) was sputtered using an ITO target (97 wt% In_2O_3 with 3 wt% SnO_2). On the front side, a 15-nm IZO layer (5,000 ohms per square) was sputtered using an IZO target (97 wt% In_2O_3 with 3 wt% ZnO). The front and rear transparent conductive oxide layers each had an area of $1.1 \times 1.1 \text{ cm}^2$. A 1,000-nm Ag back electrode was then evaporated on the rear ITO. The SHJ bottom cells were subsequently laser-cut into $2.50 \times 2.50\text{-cm}^2$ substrates for tandem integration.

Prior to spin-coating the perovskite, the SHJ bottom cells were annealed in air at 200 °C for 20 min, followed by ultraviolet-ozone cleaning for 5 min. In the subsequent process deposition, perovskite top cells were fabricated on the front side of SHJ bottom cells and annealed at 100 °C for 20 min. Except for the C_{60} and SnO_2 deposition steps, all other functional-layer processes are identical to those used for single-junction perovskite devices. The concentration of the perovskite precursor solution (incorporating both RbOCN and MAI as the target precursor) was increased to 1.8 M. The thickness of the C_{60} layer was reduced to 12 nm, and the number of ALD- SnO_2 cycles was decreased to 140. A 40-nm semitransparent IZO electrode

(130 ohms per square) was deposited on the ALD- SnO_2 thin films by radio-frequency magnetron sputtering at 0.4 Pa and a power of 30 W. Subsequently, an 800-nm Ag metal grid was evaporated on the IZO. Finally, a 115-nm MgF_2 anti-reflection coating was applied by thermal evaporation. The device active area of the perovskite–silicon TSCs was defined using a 1.001-cm^2 aperture mask.

Characterization

Unit-cell parameters were determined, and data were collected on an XtaLAB Synergy-i diffractometer using $\text{Mo K}\alpha$ radiation ($\lambda = 0.71073 \text{ \AA}$) in scanning mode at 100(1) K. The single-crystal structure was solved and refined using SHELXT and OLEX2^{40–42}. The surface morphology of perovskite films and cross-sectional device structure were examined by SEM (FEI Apreo LoVac). The perovskite bulk-heterojunction structure was further examined by HRTEM. The crystal structure and absorption properties of the perovskite films were analysed by X-ray diffraction (D8 Advance) and ultraviolet–visible spectroscopy (DektakXT). XPS and ultraviolet photoelectron spectroscopy measurements were performed on a Thermo K-Alpha+ system. The space-charge-limited current spectra were recorded using a commercialized system (XPQY-EQE-350-1100, Guangzhou Xi Pu Optoelectronics Technology), equipped with an integrating sphere (GPS-4P-SL, Labsphere) and a photodetector array (S7031-1006, Hamamatsu Photonics). Steady-state PL and TRPL spectra of perovskite films were recorded using an Edinburgh Instruments FLS980 fluorescence spectrometer with a 460-nm excitation source. The contact angle was measured using a DSA100 KRÜSS Drop Shape Analyzer, and the ADVANCE software was used to obtain the image and angle information. Elemental depth profiling of perovskite films was performed using time-of-flight secondary ion mass spectrometry (ION-TOF). The performance of all unencapsulated solar cells was measured under ambient conditions (35–45% relative humidity and 25–30 °C temperature). J – V characteristics and photovoltaic parameters were measured using a Newport solar simulator (Keithley series 2400, ORIEL-SOI3A) under AM1.5G illumination at 100 mW cm^{-2} , calibrated with a silicon reference cell. The EQE spectra were recorded by using a QE-R instrument from Enlitech, calibrated with a silicon reference cell. For the top perovskite subcell, the tandem device was preconditioned under 850 nm near-infrared bias illumination at a 0.5 V bias voltage. The bottom silicon subcell, meanwhile, was preconditioned under 550 nm blue bias light while also held at a 0.5 V bias voltage.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Crystallographic data for the structures reported in this paper have been deposited at the Cambridge Crystallographic Data Centre (CCDC) under the deposition number CCDC 2506067, corresponding to RbPbI_2OCN . Copies of the data can be obtained free of charge via <https://www.ccdc.cam.ac.uk/structures/>. Source data are provided with this paper.

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Author contributions

Yousheng Wang, J.F., Y.M. and Q.M. conceived of the idea. Q.M., J.F. and Yousheng Wang carried out the experimental work and data analysis. J.F., M.L., Y.Y. and H.L. assisted with theoretical calculations. Q.M., Yong Wang, C.H., J.Z., Y.P., D.X. and J.P. contributed to material and device characterization, including measurements by SEM, PL, TRPL, XPS and tandem device evaluation. The paper was drafted by Q.M., Yousheng Wang and J.F. and subsequently revised and edited with contributions from Yousheng Wang, J.F, Z.L. and Y.M. Yousheng Wang, J.F, Z.L. and Y.M. supervised this project. All authors have approved the final version of the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	Black masks with aperture areas of 0.09 cm ² and 1.001 cm ² were used to the tested areas for the single-junction and perovskite/Si tandem cells, respectively. Explain why this information is not reported/not relevant.
Method used to determine the device area	☒ Yes ☐ No	See the Methods section. Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	See Fig. 3g, Fig. 4c-d, Supplementary Figure 35a, and Supplementary Figure 37
Voltage scan conditions	☒ Yes ☐ No	The single-junction cells were measured with a scanning speed of 0.13 V/s and dwell time of 1 ms from 1.30 V to -0.1 V (reverse scan) and -0.1 to 1.30 V (forward scan). The perovskite/silicon tandem cells were measured with step of 0.025V, delay time of 0.01s from 2.1 V to -0.02 V (reverse scan) and -0.02 to 2.1 V (forward scan). Explain why this information is not reported/not relevant.
Test environment	☒ Yes ☐ No	All solar cells were tested under AM 1.5G illumination at ambient-air conditions (25° C, 40-50% R.H.) Explain why this information is not reported/not relevant.
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	Provide a description of the protocol. No preconditioning was used in this work.
Stability of the J-V characteristic	☒ Yes ☐ No	See Figure 4f. Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	No unusual behaviour was found. Explain why this information is not reported/not relevant.
Related experimental data	☒ Yes ☐ No	See Fig. 3g, Fig. 4c-d, Supplementary Figure 35a, and Supplementary Figure 37 Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	See Fig. 3i and Fig. 4e. Explain why this information is not reported/not relevant.
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A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	See the main text and Fig. 3i and Fig. 4e. Explain why this information is not reported/not relevant.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☒ Yes ☐ No	See Methods section. Explain why this information is not reported/not relevant.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Solar cells were tested under Xenon lamp solar simulator and AM1.5G. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The reference cell was calibrated and certified by an accredited China Photovoltaic Testing Center (CPVT). Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	The spectrum used for measurement matches well with the reference cell, and tandem cell was certified by the China Photovoltaic Testing Center (CPVT). Explain why this information is not reported/not relevant.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	0.09 cm² and 1.001 cm² for single-junction and perovskite/Si tandem cells, respectively. Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☒ Yes ☐ No	All tested cells were measured using identical masks. Explain why this information is not reported/not relevant.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	One tandem cell was send out for certification by the China Photovoltaic Testing Center (CPVT). Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	See Supplementary S37. Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	10 cells for both single-junction and tandem cells. Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	See Fig. 3h and Fig. 4b. Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Long-term operational stability of the encapsulated tandem cells under MPPT at AM 1.5G illumination in ambient-air conditions under the ISOS-L-1 protocol. Outdoor stability test under the ISOS-O-2 protocol, See Fig. 4g-h, and Methods section. Explain why this information is not reported/not relevant.